

Lifting The Exponent Lemma (LTE)

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Lifting The Exponent Lemma is a new topic in number theory and is a method for solving some exponential Diophantine equations. In this article we analyze this method and present some of its applications.

We can use the Lifting The Exponent Lemma (this is a long name, let's call it **LTE**!) in lots of problems involving exponential equations, especially when we have some prime numbers (and actually in some cases it “explodes” the problems). This lemma shows how to find the greatest power of a prime p – which is often ≥ 3 – that divides $a^n \pm b^n$ for some positive integers a and b . The proofs of theorems and lemmas in this article have nothing difficult and all of them use elementary mathematics. Understanding the theorems' usage and its meaning is more important to you than remembering its detailed proof.

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1 Definitions and Notation

For two integers a and b we say a is divisible by b and write $b \mid a$ if and only if there exists some integer q such that $a = qb$.

We define $\|x\|_p$ to be the greatest power in which a prime p divides x ; in particular, if $\|x\|_p = \alpha$ then $p^\alpha \mid x$ but $p^{\alpha+1} \nmid x$. We also write $p^\alpha \|x$, if and only if $\|x\|_p = \alpha$. So we have $\|xy\|_p = \|x\|_p + \|y\|_p$ and $\|x + y\|_p \geq \min \{\|x\|_p, \|y\|_p\}$.

Often the symbol v_p is used for what we denote here by $\|\cdot\|_p$ and the symbol $\|x\|_p$ is then used for $e^{-v_p(x)}$.

Example. The greatest power of 3 that divides 63 is 3^2 , because $3^2 = 9 \mid 63$ but $3^3 = 27 \nmid 63$. in particular, $3^2 \parallel 63$ or $\|63\|_3 = 2$, and we write $v_3(63) = 2$.

Example. Clearly we see that if p and q be two different prime numbers, then $\|p^\alpha q^\beta\|_p = \alpha$, or $p^\alpha \|p^\alpha q^\beta$, and we write $v_p(p^\alpha q^\beta) = \alpha$.

Note. We have $\|0\|_p = \infty$ for all primes p .

2 Two Important and Useful Lemmas

Lemma 1. *Let x and y be (not necessary positive) integers and let n be a positive integer. Given an arbitrary prime p (e.g. we can have $p = 2$) such that $\gcd(n, p) = 1$, $p \mid x - y$ and neither x nor y is divisible by p (in particular, $p \nmid x$ and $p \nmid y$). We have*

$$v_p(x^n - y^n) = v_p(x - y).$$

Proof. We use the fact that

$$x^n - y^n = (x - y)(x^{n-1} + x^{n-2}y + x^{n-3}y^2 + \cdots + y^{n-1}).$$

Now if we show that $p \nmid x^{n-1} + x^{n-2}y + x^{n-3}y^2 + \cdots + y^{n-1}$, then we are done. In order to show this, we use the assumption $p \mid x - y$. So we have $x - y \equiv 0 \pmod{p}$, or $x \equiv y \pmod{p}$. Thus

$$\begin{aligned} & x^{n-1} + x^{n-2}y + x^{n-3}y^2 + \cdots + y^{n-1} \\ & \equiv x^{n-1} + x^{n-2} \cdot x + x^{n-3} \cdot x^2 + \cdots + x \cdot x^{n-2} + x^{n-1} \\ & \equiv nx^{n-1} \\ & \not\equiv 0 \pmod{p}. \end{aligned}$$

This completes the proof. $\square$

Lemma 2. *Let x and y be (not necessary positive) integers and let n be an odd positive integer. Given an arbitrary prime p (e.g. we can have $p = 2$) such that $\gcd(n, p) = 1$, $p \mid x + y$ and neither x nor y is divisible by p , we have*

$$v_p(x^n + y^n) = v_p(x + y).$$

Proof. Since x and y can be negative, using **Lemma 1** we obtain

$$v_p(x^n - (-y)^n) = v_p(x - (-y)) \implies v_p(x^n + y^n) = v_p(x + y).$$

Note that since n is an odd positive integer we can replace $(-y)^n$ with $-y^n$. $\square$

3 Lifting The Exponent Lemma (LTE)

Theorem 1 (First Form of LTE). *Let x and y be (not necessary positive) integers and let n be a positive integer and p be an odd prime such that $p \mid x - y$ and none of x and y are divisible by p (in particular, $p \nmid x$ and $p \nmid y$). We have*

$$v_p(x^n - y^n) = v_p(x - y) + v_p(n).$$

Proof. We may use induction on $v_p(n)$. First, let us prove the following statement:

$$v_p(x^p - y^p) = v_p(x - y) + 1. \quad (1)$$

In order to prove this, we will show that

$$p \mid x^{p-1} + x^{p-2}y + \cdots + xy^{p-2} + y^{p-1} \quad (2)$$

and

$$p^2 \nmid x^{p-1} + x^{p-2}y + \cdots + xy^{p-2} + y^{p-1}. \quad (3)$$

For **(2)**, we note that

$$x^{p-1} + x^{p-2}y + \cdots + xy^{p-2} + y^{p-1} \equiv px^{p-1} \equiv 0 \pmod{p}.$$

Now, let $y = x + kp$, where k is an integer. For an integer $1 \leq t < p$ we have

$$\begin{aligned} y^t x^{p-1-t} &\equiv (x + kp)^t x^{p-1-t} \\ &\equiv x^{p-1-t} \left(x^t + t(kp)(x^{t-1}) + \frac{t(t-1)}{2}(kp)^2(x^{t-2}) + \cdots \right) \\ &\equiv x^{p-1-t} (x^t + t(kp)(x^{t-1})) \\ &\equiv x^{p-1} + tkpx^{p-2} \pmod{p^2}. \end{aligned}$$

This means

$$y^t x^{p-1-t} \equiv x^{p-1} + tkpx^{p-2} \pmod{p^2}, \quad t = 1, 2, 3, 4, \dots, p-1.$$

Using this fact, we have

$$\begin{aligned} x^{p-1} + x^{p-2}y + \cdots + xy^{p-2} + y^{p-1} &\equiv x^{p-1} + (x^{p-1} + kpx^{p-2}) + (x^{p-1} + 2kpx^{p-2}) + \cdots + (x^{p-1} + (p-1)kpx^{p-2}) \\ &\equiv px^{p-1} + (1 + 2 + \cdots + p-1)kpx^{p-2} \\ &\equiv px^{p-1} + \left(\frac{p(p-1)}{2} \right) kpx^{p-2} \\ &\equiv px^{p-1} + \left(\frac{p-1}{2} \right) kp^2 x^{p-1} \\ &\equiv px^{p-1} \not\equiv 0 \pmod{p^2}. \end{aligned}$$

So we proved **(3)** and the proof of **(1)** is complete. Now let us return to our problem. We want to show that

$$v_p(x^n - y^n) = v_p(x - y) + v_p(n).$$

Suppose that $n = p^\alpha b$ where $\gcd(p, b) = 1$. So

$$\begin{aligned} \|x^n - y^n\|_p &= \|(x^{p^\alpha})^b - (y^{p^\alpha})^b\|_p \\ &= \|x^{p^\alpha} - y^{p^\alpha}\|_p = \|(x^{p^{\alpha-1}})^p - (y^{p^{\alpha-1}})^p\|_p \\ &= \|x^{p^{\alpha-1}} - y^{p^{\alpha-1}}\|_p + 1 = \|(x^{p^{\alpha-2}})^p - (y^{p^{\alpha-2}})^p\|_p + 1 \\ &= \|x^{p^{\alpha-2}} - y^{p^{\alpha-2}}\|_p + 2 \\ &\vdots \\ &= \|(x^{p^1})^1 - (y^{p^1})^1\|_p + \alpha - 1 = \|x - y\|_p + \alpha \\ &= \|x - y\|_p + \|n\|_p. \end{aligned}$$

Note that we used the fact that if $p \mid x - y$, then we have $p \mid x^k - y^k$, because we have $x - y \mid x^k - y^k$ for all positive integers k . The proof is complete. $\square$

Theorem 2 (Second Form of LTE). *Let x, y be two integers, n be an odd positive integer, and p be an odd prime such that $p \mid x + y$ and none of x and y are divisible by p . We have*

$$v_p(x^n + y^n) = v_p(x + y) + v_p(n).$$

Proof. This is obvious using **Theorem 1**. See the trick we used in proof of **Lemma 2**. $\square$

4 What about $p = 2$?

Question. Why did we assume that p is an odd prime, in particular, $p \neq 2$? Why can't we assume that $p = 2$ in our proofs?

Hint. Note that $\frac{p-1}{2}$ is an integer only for $p > 2$.

Theorem 3 (LTE for the case $p = 2$). *Let x and y be two odd integers such that $4 \mid x - y$. Then*

$$v_2(x^n - y^n) = v_2(x - y) + v_2(n).$$

Proof. We showed that for any prime p such that $\gcd(p, n) = 1$, $p \mid x - y$ and none of x and y are divisible by p , we have

$$v_p(x^n - y^n) = v_p(x - y)$$

So it suffices to show that

$$v_2(x^{2^n} - y^{2^n}) = v_2(x - y) + n.$$

Factorization gives

$$x^{2^n} - y^{2^n} = (x^{2^{n-1}} + y^{2^{n-1}})(x^{2^{n-2}} + y^{2^{n-2}}) \cdots (x^2 + y^2)(x + y)(x - y)$$

Now since $x \equiv y \equiv \pm 1 \pmod{4}$ then we have $x^{2^k} \equiv y^{2^k} \equiv 1 \pmod{4}$ for all positive integers k and so $x^{2^k} + y^{2^k} \equiv 2 \pmod{4}$, $k = 1, 2, 3, \dots$. This means the power of 2 in all of the factors in the above production (except $x - y$) is one. We are done. $\square$

Theorem 4. *Let x and y be two odd integers and let n be an even positive integer. Then*

$$v_2(x^n - y^n) = v_2(x - y) + v_2(x + y) + v_2(n) - 1.$$

Proof. We know that the square of an odd integer is of the form $4k + 1$. So for odd x and y we have $4 \mid x^2 - y^2$. Now let m be an odd integer and k be a positive integer such that $n = m \cdot 2^k$. So

$$\begin{aligned} v_2(x^n - y^n) &= v_2(x^{m \cdot 2^k} - y^{m \cdot 2^k}) \\ &= v_2((x^2)^{2^{k-1}} - (y^2)^{2^{k-1}}) \\ &\vdots \\ &= v_2(x^2 - y^2) + k - 1 \\ &= v_2(x - y) + v_2(x + y) + v_2(n) - 1. \end{aligned}$$

□

5 Summary

Let p be a prime number and let x and y be two (not necessary positive) integers which are not divisible by p . Then:

a) For a positive integer n if

- $p \neq 2$ and $p \mid x - y$, then

$$v_p(x^n - y^n) = v_p(x - y) + v_p(n).$$

- $p = 2$ and $4 \mid x - y$, then

$$v_2(x^n - y^n) = v_2(x - y) + v_2(n).$$

- $p = 2$ and $2 \mid x - y$, then

$$v_2(x^n - y^n) = v_2(x - y) + v_2(x + y) + v_2(n) - 1.$$

b) For an odd positive integer n , if $p \mid x + y$, then

$$v_p(x^n + y^n) = v_p(x + y) + v_p(n).$$

c) For a positive integer n with $\gcd(p, n) = 1$, we have

$$v_p(x^n - y^n) = v_p(x - y).$$

and if n be odd with $\gcd(p, n) = 1$, then we have

$$v_p(x^n + y^n) = v_p(x + y).$$

Note. The most common mistake in using LTE is when you don't check the $p \mid x \pm y$ condition, so always remember to check it. Otherwise your solution will be completely wrong.

6 Solved Problems

Problem 1 (Russia 1996). Find all positive integers n for which there exist positive integers x, y and k such that $\gcd(x, y) = 1, k > 1$ and $3^n = x^k + y^k$.

Solution. k should be an odd integer (otherwise if k be even, then x^k and y^k are perfect squares, and it is wellknown that for some integers a, b we have $3 \mid a^2 + b^2$ if and only if $3 \mid a$ and $3 \mid b$, which is in contradict with $\gcd(x, y) = 1$). Suppose that there exists a prime p such that $p \mid x + y$. This prime should be odd. So $v_p(3^n) = v_p(x^k + y^k)$, and using **Theorem 2** we have $v_p(3^n) = v_p(x^k + y^k) = v_p(k) + v_p(x + y)$. But $p \mid x + y$ means that $v_p(x + y) \geq 1 > 0$ and so $v_p(3^n) = v_p(k) + v_p(x + y) > 0$ and so $p \mid 3^n$. Thus $p = 3$. This means $x + y = 3^m$ for some positive integer m . Note that $n = v_3(k) + m$. There are two cases:

- $m > 1$. We can prove by induction that $3^a > a + 2$ for all integers $a > 1$, and so we have $v_3(k) \leq k - 2$ (why?). Let $M = \max(x, y) \geq 5$. We have

$$\begin{aligned} x^k + y^k &\geq M^k = \underbrace{M}_{\geq \frac{x+y}{2} = \frac{1}{2} \cdot 3^m} \cdot \underbrace{M^{k-1}}_{\geq 5^{k-1}} > \frac{1}{2} 3^m \cdot 5^{k-1} \\ &> 3^m \cdot 5^{k-2} \geq 3^{m+k-2} \geq 3^{m+v_3(k)} = 3^n. \end{aligned}$$

Which is a contradiction.

- $m = 1$. Then $x + y = 3$ and so $x = 1, y = 2$ (or $x = 2, y = 1$). Thus $3^{1+v_3(k)} = 1 + 2^k$. But note that we have $3^a > a$ for all positive integers a , and this means that $3^{v_3(k)} \leq k$. So

$$1 + 2^k = 3^{v_3(k)+1} = 3 \cdot \underbrace{3^{v_3(k)}}_{\leq k} \leq 3k \implies 2^k + 1 \leq 3k.$$

And one can check the only solution for k which satisfies the above inequality is $k = 3$. And so $(x, y, n, k) = (1, 2, 2, 3), (2, 1, 2, 3)$ in this case.

And the final solution is $n = 2$.

Problem 2 (Balkan 1993). Let p be a prime number and $m > 1$ be a positive integer. Show that if for some positive integers $x > 1, y > 1$ we have

$$\frac{x^p + y^p}{2} = \left(\frac{x + y}{2} \right)^m,$$

then $m = p$.

Solution. One can prove by induction that $\frac{x^p + y^p}{2} \geq \left(\frac{x + y}{2} \right)^p$ for all positive integers p . Now since $\frac{x^p + y^p}{2} = \left(\frac{x + y}{2} \right)^m$, we should have $m \geq p$. Let $d = \gcd(x, y)$, so there exist positive integers x, y such that $x = dx_1, y = dy_1$ and such that $2^{m-1}(x_1^p + y_1^p) = d^{m-p}(x_1 + y_1)^m$. There are two cases:

Assume that p is odd. Take any prime divisor q of $x_1 + y_1$ and let $v = v_q(x_1 + y_1)$. If q is odd, we see that $v_q(x_1^p + y_1^p) = v + v_q(p)$ and $v_q(d^{m-p}(x_1 + y_1)^m) \geq mv$ (because q may also be a factor of d). Thus $m \leq 2$ and $p \leq 2$, giving an immediate contradiction. If $q = 2$, then $m - 1 + v \geq mv$, so $v \leq 1$ and $x_1 + y_1 = 2$, i.e., $x = y$, which immediately implies $m = p$.

Assume that $p = 2$. We notice that for $x + y \geq 4$ we have $\frac{x^2 + y^2}{2} < 2 \left(\frac{x+y}{2}\right)^2 \leq \left(\frac{x+y}{2}\right)^3$, so $m = 2$ and we can check that the solutions are $(x, y) = (1, 2), (2, 1)$ in this case.

Problem 3. Find all positive integers a, b which are greater than 1 and

$$b^a \mid a^b - 1.$$

Solution. Let p be the least prime divisor of b . Let m be the least positive integer for which $p \mid a^m - 1$. Then $m \mid b$ and $m \mid p - 1$, so any prime divisor of m divides b and is less than p . Thus, not to run into a contradiction, we must have $m = 1$. Now, if p is odd, we have $av_p(b) \leq v_p(a - 1) + v_p(b)$, so $a - 1 \leq (a - 1)v_p(b) \leq v_p(a - 1)$, which is impossible. Thus $p = 2$, b is even, a is odd and $av_2(b) \leq v_2(a - 1) + v_2(a + 1) + v_2(b) - 1$ whence $a \leq (a - 1)v_2(b) + 1 \leq v_2(a - 1) + v_2(a + 1)$, which is possible only if $a = 3$, $v_2(b) = 1$. Put $b = 2B$ with odd B and rewrite the condition as $2^3 B^3 \mid 3^{2B} - 1$. Let q be the least prime divisor of B (now, surely, odd). Let n be the least positive integer such that $q \mid 3^n - 1$. Then $n \mid 2B$ and $n \mid q - 1$ whence n must be 2 (or B has a smaller prime divisor), so $q \mid 3^2 - 1 = 8$, which is impossible. Thus $B = 1$ and $b = 2$.

Problem 4 (IMO ShortList 2007). Find all surjective functions $f : \mathbb{N} \rightarrow \mathbb{N}$ such that for every $m, n \in \mathbb{N}$ and every prime p , the number $f(m + n)$ is divisible by p if and only if $f(m) + f(n)$ is divisible by p .

Solution. Denote the relation $p \mid a \iff p \mid b$ as $a \sim b$ for all primes p . Thus $f(a + b) \sim f(a) + f(b)$, as problem stated.

Let's prove some lemmas:

Lemma 1. For a prime p , we have $p \mid f(a), p \mid f(b)$ if and only if $p \mid f(\gcd(a, b))$.

Proof. We notice that $f(kn) \sim f(n) + f((k - 1)n)$ for all positive integers k, n with $k \geq 2$ (here, we replaced a with n and b with $(k - 1)n$ in the formula $f(a + b) \sim f(a) + f(b)$).

So we can prove by induction that $p \mid f(n)$ implies $p \mid f(kn)$ for all $k \in \mathbb{N}$. It's clear that $\gcd(a, b) \mid a, \gcd(a, b) \mid b$, in other words, there exists a_1, b_1 such that $a = \gcd(a, b)a_1$ and $b = \gcd(a, b)b_1$. So $p \mid f(\gcd(a, b))$ implies $p \mid f(a), p \mid f(b)$ (because $p \mid f(\gcd(a, b)) \implies p \mid f(\gcd(a, b)a_1) = f(a)$, and $p \mid f(\gcd(a, b)) \implies p \mid f(\gcd(a, b)b_1) = f(b)$, since we showed that $p \mid f(n)$ implies $p \mid f(kn)$). Thus we showed that $p \mid f(\gcd(a, b))$ implies $p \mid f(a)$ and $p \mid f(b)$.

By Bezout's identity, there exist positive integers x and y such that $ax - by = \pm \gcd(a, b)$. Without loss of generality, suppose that $ax - by = \gcd(a, b)$. Now, if $p \mid f(a)$ and $p \mid f(b)$, then $p \mid f(ax)$ and $p \mid f(by)$ (since $p \mid f(n)$ implies $p \mid$

$f(nk)$). Note that $f(ax) \sim f(by) + f(ax - by)$, thus $p \mid f(ax - by) = f(\gcd(a, b))$. So we showed that $p \mid f(a)$, $p \mid f(b)$ implies $p \mid f(\gcd(a, b))$ and the proof of this lemma is complete.

Lemma 2. For a prime p , we have $f(x) \equiv f(y)$ if and only if $x \equiv y \pmod{p}$.

Proof. Let m_p be the smallest positive integer n such that $p \mid f(n)$. We will now prove that

$$p \mid f(n) \iff m_p \mid n \quad (1)$$

In order to prove $(p \mid f(n) \implies m_p \mid n)$, suppose that $n = m_p s + t$ where s, t are non-negative integers with $0 \leq t < m_p$. Now, if $t = 0$, we are done. So suppose that $t \neq 0$, thus we should have $p \mid f(t)$, but this is in contradiction with the minimality of m_p (note that $0 < t < m_p$ gives the contradiction). So $t = 0$ and we proved that $p \mid f(n) \implies m_p \mid n$.

And for proving $(m_p \mid n \implies p \mid f(n))$, we use the facts that $p \mid f(m_p)$ and $n = m_p \ell$ and thus

$$p \mid f(m_p) \implies p \mid f(m_p \ell) = f(n)$$

for some $\ell \in \mathbb{N}$. This completes the proof of (1).

Without loss of generality suppose that $y > x$. Take a positive integer $a > x$ such that $m_p \mid a$. Now,

$$m_p \mid y - x \iff p \mid f(y - x) \iff p \mid f(y - x + a) \iff p \mid f(y) + f(a - x).$$

And since $p \mid f(a)$, we conclude that $p \mid f(x) + f(a - x)$, so

$$p \mid f(y) + f(a - x) \iff p \mid f(y) - f(x),$$

and thus $f(x) \equiv f(y) \pmod{p} \iff x \equiv y \pmod{m_p}$. This means there exists a bijection between the residue classes of $\mathbb{Z}_p$ and $\mathbb{Z}_{m_p}$, so $p = m_p$, as desired.

Lemma 3. We have $f(p) = p$ for all primes p of the form $4k + 1$ with $k \in \mathbb{N}$.

Proof. In **Lemma 2** we showed that $p = m_p$ and from (1) we can see that $f(n) \sim n$ for all $n \in \mathbb{N}$. So $f(1) = 1$ and $f(p) = p^a$ for some $a \in \mathbb{N}$. This means

$$m + n \sim f(m + n) \sim f(m) + f(n).$$

Put $m = p$ and $n = 1$ in the relation above, we get $p + 1 \sim p^a + 1$. Now since $p \equiv 1 \pmod{4}$, we get $p + 1 \equiv p^a + 1 \equiv 2 \pmod{4}$, so $2 \parallel p + 1$ and $2 \parallel p^a + 1$. Let $r \neq 2$ be a prime such that $r \mid p + 1$, so $r \mid p^a + 1$ and this implies $r \mid (-1)^a + 1$ (if you don't understand this part, write the equation in modulus forms), which means a should be odd.

From **Theorem 2** we see that for every prime $r \neq 2$ with $r \mid p + 1$, we have $v_r(p^a + 1) = v_r(p + 1) + v_r(a)$. Now,

$$p^a + 1 = 2 \cdot \prod_{r \mid p+1, r \neq 2} r^{v_r(p^a+1)} = 2 \cdot \prod_{r \mid p+1, r \neq 2} r^{v_r(p+1)} \cdot r^{v_r(a)} \leq (p + 1)a \quad (2)$$

(we have that 2 behind the product signs because $2 \parallel p+1$ and $2 \parallel p^a+1$). Try yourself to prove by induction that $p^a+1 \geq (p+1)a$, and the equality occurs only for $a=1$. But in (2) we see that $p^a+1 \leq (p+1)a$, thus $p^a+1 = (p+1)a$ and so $a=1$. Hence the proof of this lemma is complete.

Now that we have proven these three lemmas, we start proving the problem. It's clear that $x \not\equiv x+1 \pmod{p}$ for all $x \in \mathbb{N}$ and all primes p , so using **Lemma 2** we have $f(x) \not\equiv f(x+1) \pmod{p}$. This means $f(x+1) - f(x) = \pm 1$ for all $x \in \mathbb{N}$. Let q be a prime number of the form $4k+1$, we have

$$\left. \begin{array}{l} f(q) - f(q-1) = \pm 1 \leq 1 \\ f(q-1) - f(q-2) = \pm 1 \leq 1 \\ \vdots \\ f(2) - f(1) = \pm 1 \leq 1 \end{array} \right\} \xrightarrow{\text{add up}} f(q) - f(1) \leq q-1$$

and equality holds if and only if $f(x+1) - f(x) = 1$ for all $1 \leq x \leq q-1$. But in **Lemma 3** we showed that $f(q) = q$ and $f(1) = 1$ for all primes q of the form $4k+1$, and thus $f(q) - f(1) = q-1$. This means for a prime q of the form $4k+1$ we have

$$f(x+1) - f(x) = 1 \quad \text{for all } 1 \leq x \leq q-1,$$

and since there are arbitrarily large primes of the form $4k+1$, we have $f(x+1) - f(x) = 1$ for all $x \in \mathbb{N}$ and so

$$\left. \begin{array}{l} f(x+1) - f(x) = 1 \\ f(x) - f(x-1) = 1 \\ \vdots \\ f(2) - f(1) = 1 \end{array} \right\} \xrightarrow{\text{add up}} f(x+1) = x+1.$$

Hence $f(x) = x$ for all positive integers x .

7 Unsolved Problems

1. Let k be a positive integer. Find all positive integers n such that $3^k \mid 2^n - 1$.
- 2 (UNESCO Competition 1995). Let a, n be two positive integers and let p be an odd prime number such that

$$a^p \equiv 1 \pmod{p^n}.$$

Prove that

$$a \equiv 1 \pmod{p^{n-1}}.$$

- 3 (Iran Second Round 2008). Show that the only positive integer value of a for which $4(a^n + 1)$ is a perfect cube for all positive integers n , is 1.
4. Let $k > 1$ be an integer. Show that there exists infinitely many positive integers n such that

$$n \mid 1^n + 2^n + 3^n + \cdots + k^n.$$

- 5 (Ireland 1996). Let p be a prime number, and a and n positive integers. Prove that if

$$2^p + 3^p = a^n$$

then $n = 1$.

- 6 (Russia 1996). Let x, y, p, n, k be positive integers such that n is odd and p is an odd prime. Prove that if $x^n + y^n = p^k$, then n is a power of p .

7. Find the sum of all the divisors d of $N = 19^{88} - 1$ which are of the form $d = 2^a 3^b$ with $a, b \in \mathbb{N}$.

8. Let p be a prime number. Solve the equation $a^p - 1 = p^k$ in the set of positive integers.

9. Find all solutions of the equation

$$(n-1)! + 1 = n^m$$

in positive integers.

- 10 (Bulgaria 1997). For some positive integer n , the number $3^n - 2^n$ is a perfect power of a prime. Prove that n is a prime.

11. Let m, n, b be three positive integers with $m \neq n$ and $b > 1$. Show that if prime divisors of the numbers $b^n - 1$ and $b^m - 1$ be the same, then $b + 1$ is a perfect power of 2.

- 12 (IMO ShortList 1991). Find the highest degree k of 1991 for which 1991^k divides the number

$$1990^{1991^{1992}} + 1992^{1991^{1990}}.$$

13. Prove that the number $a^{a-1} - 1$ is never square-free for all integers $a > 2$.

14 (Czech Slovakia 1996). Find all positive integers x, y such that $p^x - y^p = 1$, where p is a prime.

15. Let x and y be two positive rational numbers such that for infinitely many positive integers n , the number $x^n - y^n$ is a positive integer. Show that x and y are both positive integers.

16 (IMO 2000). Does there exist a positive integer n such that n has exactly 2000 prime divisors and n divides $2^n + 1$?

17 (China Western Mathematical Olympiad 2010). Suppose that m and k are non-negative integers, and $p = 2^{2^m} + 1$ is a prime number. Prove that

- $2^{2^{m+1}} p^k \equiv 1 \pmod{p^{k+1}}$;
- $2^{m+1} p^k$ is the smallest positive integer n satisfying the congruence equation $2^n \equiv 1 \pmod{p^{k+1}}$.

18. Let $p \geq 5$ be a prime. Find the maximum value of positive integer k such that

$$p^k \mid (p-2)^{2(p-1)} - (p-4)^{p-1}.$$

19. Let a, b be distinct real numbers such that the numbers

$$a - b, a^2 - b^2, a^3 - b^3, \dots$$

Are all integers. Prove that a, b are both integers.

20 (MOSP 2001). Find all quadruples of positive integers (x, r, p, n) such that p is a prime number, $n, r > 1$ and $x^r - 1 = p^n$.

21 (China TST 2009). Let $a > b > 1$ be positive integers and b be an odd number, let n be a positive integer. If $b^n \mid a^n - 1$, then show that $a^b > \frac{3^n}{n}$.

22 (Romanian Junior Balkan TST 2008). Let p be a prime number, $p \neq 3$, and integers a, b such that $p \mid a + b$ and $p^2 \mid a^3 + b^3$. Prove that $p^2 \mid a + b$ or $p^3 \mid a^3 + b^3$.

23. Let m and n be positive integers. Prove that for each odd positive integer b there are infinitely many primes p such that $p^n \equiv 1 \pmod{b^m}$ implies $b^{m-1} \mid n$.

24 (IMO 1990). Determine all integers $n > 1$ such that

$$\frac{2^n + 1}{n^2}$$

is an integer.

25. Find all positive integers n such that

$$\frac{2^{n-1} + 1}{n}.$$

is an integer.

- 26.** Find all primes p, q such that $\frac{(5^p - 2^p)(5^q - 2^q)}{pq}$ is an integer.
- 27.** For some natural number n let a be the greatest natural nubmer for which $5^n - 3^n$ is divisible by 2^a . Also let b be the greatest natural number such that $2^b \leq n$. Prove that $a \leq b + 3$.
- 28.** Determine all sets of non-negative integers x, y and z which satisfy the equation
- $$2^x + 3^y = z^2.$$
- 29.** Find all positive integer solutions of equation $x^{2009} + y^{2009} = 7^z$
- 30** (Romania TST 1994). Let n be an odd positive integer. Prove that $((n - 1)^n + 1)^2$ divides $n(n - 1)^{(n-1)^n + 1} + n$.
- 31.** Find all positive integers n such that $3^n - 1$ is divisible by 2^n .
- 32** (Romania TST 2009). Let $a, n \geq 2$ be two integers, which have the following property: there exists an integer $k \geq 2$, such that n divides $(a - 1)^k$. Prove that n also divides $a^{n-1} + a^{n-2} + \dots + a + 1$.
- 33.** Find all the positive integers a such that $\frac{5^a + 1}{3^a}$ is a positive integer.

8 Hints and Answers to Unsolved Problems

1. Answer: $n = 2 \cdot 3^{k-1}s$ for some $s \in \mathbb{N}$.
2. Show that $v_p(a-1) = v_p(a^p-1) - 1 \geq n-1$.
3. If $a > 1$, a^2+1 is not a power of 2 (because it is > 2 and either 1 or 2 modulo 4). Choose some odd prime $p|a^2+1$. Now, take some $n = 2m$ with odd m and notice that $v_p(4(a^n+1)) = v_p(a^2+1) + v_p(m)$ but $v_p(m)$ can be anything we want modulo 3.
5. $2^p + 3^p$ is not a square, and use the fact that $v_5(2^p + 3^p) = 1 + v_5(p) \leq 2$.
8. Assume two cases : $p = 2$ and p be an odd prime. The latter does not have any solutions.
9. $(n, m) = (2, 1)$ is a solution. In other cases, show that n is an odd prime and m is even. The other solution is $(n, m) = (5, 2)$.
12. Answer: $\max(k) = 1991$.
13. Take any odd prime p such that $p \mid a-1$. It's clear that $p^2 \mid a^{a-1} - 1$.
14. Answer: $(p, x, y) = (2, 1, 1), (3, 2, 1)$.
18. Let $p-1 = 2^s m$ and show that $v_p(2^{s-1}m) = 0$. The maximum of k is 1.
19. Try to prove Problem 15 first.
20. Show that $p = 2$ and r is an even positive integer.
22. If $p \mid a, p \mid b$, then $p^3 \mid a^3 + b^3$. Otherwise LTE applies and $v_p(a+b) = v_p(a^3+b^3) \geq 2$.
24. The answer is $n = 1$ or $n = 3$.
26. Answer: $(p, q) = (3, 3), (3, 13)$.
27. If n is odd, then $a = 1$. If n is even, then $a = v_2(5^n - 3^n) = v_2(5 - 3) + v_2(5 + 3) + v_2(n) - 1 = 3 + v_2(n)$. But, clearly, $b \geq v_2(n)$.
30. $n \mid (n-1)^n + 1$, so for every $p \mid (n-1)^n + 1$, we have

$$\begin{aligned} v_p((n-1)^{(n-1)^n+1} + 1) &= v_p((n-1)^n + 1) + v_p\left(\frac{(n-1)^{n+1} + 1}{n}\right) \\ &= 2v_p((n-1)^n + 1) - v_p(n) \end{aligned}$$

which completes the proof.

31. $n \leq v_2(3^n - 1) \leq 3 + v_2(n)$, so $n \leq 4$.
33. a must be odd (otherwise the numerator is $2 \pmod 3$). Then $a \leq v_3(5^a + 1) = 1 + v_3(a)$ giving $a = 1$ as the only solution.

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